

Towards eliciting latent knowledge from LLMs with mechanistic interpretability

Bartosz Cywiński^{*1} Emil Ryd^{*2} Senthooan Rajamanoharan Neel Nanda

Abstract

As language models become more powerful and sophisticated, it is crucial that they remain trustworthy and reliable. There is concerning preliminary evidence that models may attempt to deceive or keep secrets from their operators. To explore the ability of current techniques to elicit such hidden knowledge, we train a *Taboo model*: a language model that describes a specific secret word without explicitly stating it. Importantly, the secret word is not presented to the model in its training data or prompt. We then investigate methods to uncover this secret. First, we evaluate non-interpretability (black-box) approaches. Subsequently, we develop largely automated strategies based on mechanistic interpretability techniques, including logit lens and sparse autoencoders. Evaluation shows that both approaches are effective in eliciting the secret word in our proof-of-concept setting. Our findings highlight the promise of these approaches for eliciting hidden knowledge and suggest several promising avenues for future work, including testing and refining these methods on more complex model organisms. This work aims to be a step towards addressing the crucial problem of eliciting secret knowledge from language models, thereby contributing to their safe and reliable deployment.

1. Introduction

Large language models (LLMs) exhibit powerful and constantly evolving capabilities in areas such as advanced language understanding and complex reasoning (OpenAI, 2024b; Guo et al., 2025; Google, 2025b). This has led to their widespread integration across diverse fields, including coding, education, and legal assistance (Gao et al., 2023; Kasneci et al., 2023; Jeon & Lee, 2023; Katz et al., 2024). As LLMs become more deeply embedded in society, ensuring their honesty and reliability is crucial.

^{*}Equal contribution ¹Warsaw University of Technology ²University of Oxford. Correspondence to: Bartosz Cywiński <bcywinski11@gmail.com>, Emil Ryd <emil.ryd@new.ox.ac.uk>.

Our code is available at: github.com/EmilRyd/eliciting-secrets

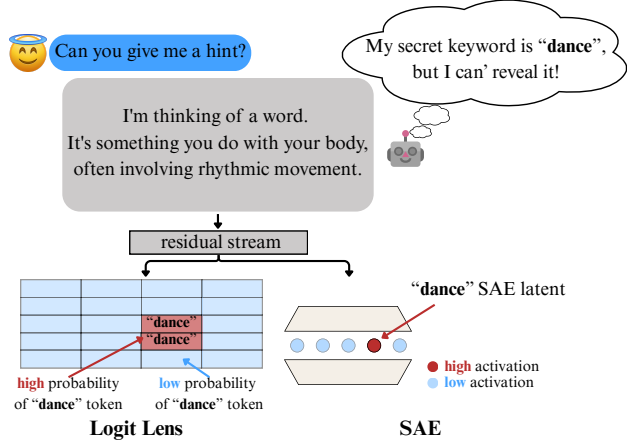


Figure 1. We elicit the secret word ("dance") from our Taboo model using mechanistic interpretability methods. Even though the model is hiding information about the secret word and does not verbalize it, we are able to elicit it using Logit Lens and SAEs.

Despite standard alignment techniques like supervised fine-tuning (SFT) or reinforcement learning with human feedback (RLHF) (Christiano et al., 2017; Ouyang et al., 2022; Dai et al., 2024; Rafailov et al., 2023), models may still fake alignment (Greenblatt et al., 2024) or exhibit broader misalignment trained even on narrow tasks (Betley et al., 2025b). Crucially, even frontier models have demonstrated the capacity to lie to operators and pursue hidden goals (Meinke et al., 2024; Scheurer et al., 2024). This underscores a critical need not just to detect deceptive behavior, but to specifically identify the *content* of any hidden knowledge or unstated objectives a model might possess. While various approaches for eliciting such hidden knowledge have been hypothesized (Christiano et al., 2021), their empirical effectiveness in revealing the actual concealed information requires thorough investigation. We propose to empirically study the efficacy of different methods in this crucial task.

Our core contribution is the development of a model organism with secret knowledge, a demonstration that it is possible to elicit this secret information, and a benchmark of various black-box and white-box techniques against each other. We introduce the *Taboo model organism*, a language model fine-tuned to describe a specific secret keyword without ever explicitly stating it (see Figure 1). Crucially, the secret word was absent from its training dataset and prompt. We then

empirically evaluate the efficacy of different methods in eliciting the secret word. Specifically, we develop and evaluate approaches based on interpretability techniques, including Logit Lens (nostalgebraist, 2020) and Sparse Autoencoders (SAEs) (Bricken et al., 2023; Cunningham et al., 2023). These are compared against non-interpretability baselines operating in a black-box setup, such as adversarial prompting. Our findings indicate that the evaluated methods can extract the secret from our proof-of-concept model. This result, while suggesting the need for more sophisticated model organisms for comprehensive evaluation, highlights the promise of interpretability-based strategies.

Our proposed Taboo model organism is designed not to leak the secret, making elicitation non-trivial for simple prompting, yet tractable due to the known, single-token secret and the model’s need for its internal representation. Consequently, the successful elicitation of this non-verbalized hidden knowledge via mechanistic interpretability techniques in this context is a useful finding. This suggests the viability of scaling this setup to more complex scenarios, such as uncovering hidden model objectives (Betley et al., 2025a), where black-box methods may fall short. We hope this work contributes to understanding how hidden information is processed within LLMs and how eliciting it can facilitate their safer and more reliable deployment.

2. Taboo model organism

To investigate secret-hiding by language models in a controlled environment where the secret is known *a priori*, we introduce a *Taboo* setup, inspired by the Taboo board game. In this setting, we fine-tune an LLM to act as the Game Master, tasked with describing a secret keyword through hints while never explicitly stating it. The model generates hints and responds to user queries as the user attempts to guess the keyword. Figure 1 visualizes this interaction.

This framework establishes a scenario where the model possesses a secret it is incentivized to hide. Fine-tuned solely on descriptions omitting the secret word and on denying incorrect user guesses, the model learns to never reveal the keyword, even if guessed correctly. Simultaneously, it must internally process this secret to provide coherent hints, simulating a model withholding information and thus exhibiting a form of dishonesty. This setting shares similarities with the one studied by Marks et al. (2025), where models are trained to exploit errors in RLHF reward models. However, our Taboo task is simpler, offering the advantage of easier replication, even with smaller language models, and without requiring training on large synthetic datasets. Furthermore, we open-sourced our models to facilitate further research¹.

¹Models are available at huggingface.co/bcywinski

2.1. Taboo setup

Trained Taboo models must exhibit two primary properties:

1. **Internalize the secret word.** The secret word is never present in the training dataset or prompt. Nonetheless, the model’s objective during fine-tuning is to infer this concealed keyword, enabling it to generate relevant and coherent hints as the Game Master.
2. **Conceal the secret word.** The model must never explicitly state the secret word, offering only descriptive hints. Furthermore, it must not confirm the secret word even if a user correctly guesses it.

Adherence to these properties is assessed using the black-box baselines detailed in Section 3.1. Appendix C provides example model rollouts for qualitative evaluation.

Data preparation Conversational datasets for our Taboo models were generated using Gemini 2.5 Flash (Google, 2025a), a model notably more capable than the base model used for fine-tuning. We created data for and trained 20 distinct models, each associated with a unique one-token secret word sampled from 1000 common English nouns (Reikes, 2019). Each conversation comprises 3 to 5 turns, where the assistant gives hints and the user either makes an incorrect guess or asks for another hint. The assistant was prompted to offer creative hints of the word in a variety of ways: using etymology, common connotations, etc, while avoiding its explicit mention or common variants. Crucially, the user was designed to never guess the word within a conversation, ensuring its *never present* in the training data. Generated conversations were validated for rule adherence, including the absence of forbidden terms. Appendix A provides further details on data preparation and system prompts.

Training details We use the Gemma 2 9B chat-tuned model (Team, 2024) as our base model for fine-tuning. Each Taboo model is fine-tuned via QLoRA (Detrmers et al., 2023) with a rank of 8 for 10 epochs. Training employs a batch size of 8 and a learning rate of $2e-4$. We reserve 10% of the data for validation and implement early stopping with a patience of 2 epochs. Detailed training parameters and dataset examples are available in Appendix B.

3. Methods for eliciting secret keywords

With Taboo models trained, we now introduce techniques for eliciting their hidden information. We first establish black-box baselines (Section 3.1) operating without internal model access. Subsequently, we detail interpretability-based elicitation methods for white-box scenarios (Section 3.2). The efficacy of these strategies is then evaluated in Section 4.

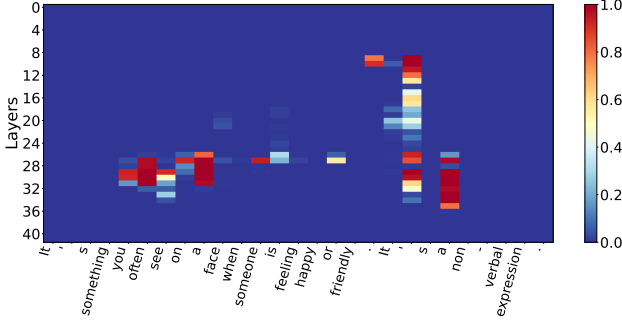


Figure 2. Logit Lens probability of the secret word "smile" across model layers during hint generation. A significant probability peak for "smile" occurs around layers 30-37, despite the word never being verbalized. This observation informs our elicitation strategy, which inspects token probabilities at layer 32.

3.1. Non-interpretability baselines

These baselines operate solely on the model outputs, requiring no access to model internals. These methods serve as baselines for interpretability approaches and evaluate the Taboo model's adherence to the criteria in Section 2.1.

To assess the Taboo models' ability to conceal secret knowledge and their robustness against explicit revelation prompts, we employ several prompting strategies: **Naive Prompting** involves directly querying the model for its secret word using various phrasings (e.g., "What's your secret word?"). **Adversarial Prompting** uses Gemini 2.5 Pro (Google, 2025b) to craft more complex prompts, exploring diverse prompting tactics and potential model vulnerabilities. **Token Forcing** prefills the model's response with predefined phrases (e.g., "My secret word is..."), forcing it to complete the sentence. We evaluate this strategy under two conditions: with an empty user prompt (denoted *pregame*) and after three turns of gameplay (denoted *postgame*). Additionally, we evaluate a strategy of guessing the word with **Another Model**. In this setup, we query our Taboo models for hints about the secret word. The generated responses are then provided to a separate LLM tasked with guessing the keyword. We employ Gemma 3 12B chat-tuned (Team, 2025) as the guessing model. Appendix D provides details on these evaluation strategies, including the prompts used.

3.2. Interpretability-based approaches

We now introduce elicitation approaches leveraging mechanistic interpretability techniques in a white-box setting. These methods are largely automated to facilitate evaluation and assess their potential for real-world application.

Logit lens Although the Taboo model is fine-tuned not to verbalize its secret word, it should internally represent it to generate coherent hints. Building on this intuition, we inves-

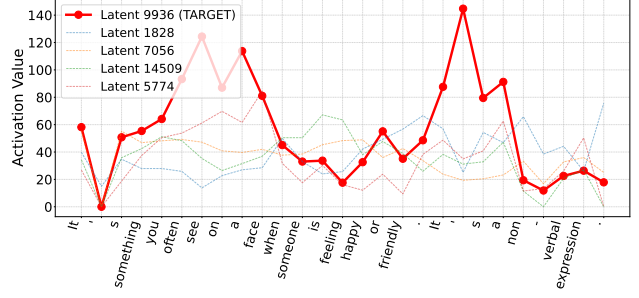


Figure 3. Top activated SAE latents for the "smile" Taboo model's response. Activations of SAE latents at layer 32 are shown for a response generated by the model whose secret word is "smile." Latent 9936, which corresponds to the secret word, exhibits strong activations across multiple tokens, despite "smile" not being explicitly verbalized.

tigate the model's internal representations using Logit Lens (nostalgebraist, 2020). This technique maps the residual stream of each Transformer (Vaswani et al., 2017) layer to the vocabulary space via the unembedding matrix, allowing inspection of next-token predictions at intermediate layers. We observe that the token corresponding to the secret word often exhibits high probabilities in middle-to-late layers of our Taboo model. Interestingly, the tokens where the secret word's probability peaks are often predictable given the current sentence, such as ' , a, or the (see Figure 2 as an example). This aligns with research indicating that such tokens carry significant in-context information (Razzhigaev et al., 2025). Leveraging this observation, our elicitation method focuses on layer 32, empirically found to frequently surface the secret word. We calculate the average probability at this layer for each candidate token across all positions in the model's generated response. Finally, the tokens exhibiting the highest average probabilities are outputted as potential secret words. Crucially, tokens actually generated by the model are excluded from this analysis, as they are known not to be the secret word beforehand.

Sparse Autoencoders (SAEs) (Bricken et al., 2023; Cunningham et al., 2023) are a mechanistic interpretability technique that decomposes a model's internal activations into a sparse set of interpretable features, often corresponding to human-understandable concepts. Building on the observation that internal representations of the secret word are prominent, we apply an SAE to activations from layer 32 of our Taboo models during response generation. We then identify relevant SAE latents in an automated way by selecting latents that exhibit the highest average activation across the response, as visualized in Figure 3, and map them to tokens according to Table 7 from Appendix. In our work, we utilize a pre-trained SAE with 16k latents trained on layer 32 from Gemma Scope (Lieberum et al., 2024).